

Design Table for *Evaluating the Claim that Preregistration and Registered Reports Restrict Exploratory Research* <https://osf.io/45qy9>

Question	Hypothesis	Sampling plan	Analysis Plan	Rationale for deciding the sensitivity of the test for confirming or disconfirming the hypothesis	Interpretation given different outcomes	Theory that could be shown wrong by the outcomes
Does registration facilitate or restrict reporting of exploratory research?	There will be a greater frequency of exploratory research in registered articles compared to traditional articles.	Based on power analysis	A 2 (registered or not) x 2 (reporting exploratory research or not) chi-square test of independence ($\alpha = .05$).	Power analysis based on range of plausible effect sizes (Cramer's $V = 0.18-0.28$), based on 80% power and $\alpha = .05$.	A positive effect would indicate that registered studies facilitate reporting of exploratory research, whereas a negative effect would indicate that they restrict reporting of exploratory research.	No theory being tested
Do Registered Reports include more frequent reports of exploratory research than preregistered or traditional articles?	There will be a greater frequency of exploratory research reported in Registered Reports compared with preregistered and traditional articles.	Based on power analysis	A 3 (registered reports, preregistered, and traditional articles) x 2 (reporting exploratory research or not) chi-square test of independence ($\alpha = .05$).	Power analysis based on range of plausible effect sizes (Cramer's $V = 0.18-0.28$), based on 80% power and $\alpha = .05$.	A positive effect would indicate that Registered Reports facilitate reporting of exploratory research over both preregistered and traditional articles. Other possibilities are that there are no differences between Registered Reports and preregistered articles and the negative effect of traditional articles including a higher rate of reporting exploratory research.	No theory being tested
Has there been change over time in reports of exploratory research in	NA – exploratory question	Based on power analysis	No inferential statistics will be used	No inferential statistics will be used	Exploratory – all possible trends in change over time open to	No theory being tested

traditional articles from 2010-2026?					interpretation.	
Has there been change over time in reports of exploratory research in registered articles from 2015-2026?	NA – exploratory question	Based on power analysis	No inferential statistics will be used	No inferential statistics will be used	Exploratory – all possible trends in change over time open to interpretation.	No theory being tested

Guidance Notes

- **Question:** articulate each research question being addressed in one sentence.
- **Hypothesis:** where applicable, a prediction arising from the research question, stated in terms of specific variables rather than concepts. Where the testability of one or more hypotheses depends on the verification of auxiliary assumptions (such as positive controls, tests of intervention fidelity, manipulation checks, or any other quality checks), any tests of such assumptions should be listed as hypotheses. Stage 1 proposals that do not seek to test hypotheses can ignore or delete this column.
- **Sampling plan:** For proposals using inferential statistics, the details of the statistical sampling plan for the specific hypothesis (e.g power analysis, Bayes Factor Design Analysis, ROPE etc). For proposals that do not use inferential statistics, include a description and justification of the sample size.
- **Analysis plan:** For hypothesis-driven studies, the specific test(s) that will confirm or disconfirm the hypothesis. For non-hypothesis-driven studies, the test(s) that will answer the research question.
- **Rationale for deciding the sensitivity of the test for confirming or disconfirming the hypothesis:** For hypothesis-driven studies that employ inferential statistics, an explanation of how the authors determined a relevant effect size for statistical power analysis, equivalence testing, Bayes factors, or other approach.
- **Interpretation given different outcomes:** A prospective interpretation of different potential outcomes, making clear which outcomes would confirm or disconfirm the hypothesis.
- **Theory that could be shown wrong by the outcomes:** Where the proposal is testing a theory, make clear what theory could be shown to be wrong, incomplete, or otherwise inadequate by the outcomes of the research.